DECISION 520Q
Section C Team 45
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Airbnb Price Prediction

Business Understanding:

Airbnb operates in a competitive hospitality and travel market where pricing is a key factor in both customer satisfaction and host profitability. The company faces the challenge of determining fair and competitive listing prices that reflect property characteristics, location, and amenities. Many listings are priced inconsistently with their true market value. Overpriced properties often experience lower booking rates and reduced host revenue, while underpriced properties limit potential earnings and distort market signals. Addressing this imbalance is critical for sustaining platform trust and improving efficiency in Airbnb's pricing system.

The business problem centers on how Airbnb can determine appropriate and competitive prices for each listing by using data-driven analysis. Specifically, the goal is to understand how factors such as location, property type, room type, and amenities influence pricing outcomes. A better understanding of these relationships will allow Airbnb to guide hosts in setting fair prices that align with market expectations and guest perceptions of value.

A data mining approach provides a precise solution to this challenge by identifying patterns and relationships within Airbnb's listing data. Using descriptive and predictive modeling, Airbnb can identify which features most strongly influence price and to what extent. Data visualization helps reveal overall trends, such as differences in pricing across cities or property types, while predictive models can estimate the expected price for a given set of characteristics. By implementing these findings into its platform, Airbnb can improve its automated pricing recommendations, allowing hosts to make more informed pricing decisions. This approach ultimately enhances booking efficiency, increases host earnings, and supports a more transparent and trustworthy marketplace.

Data Understanding:

The dataset contains more than 70,000 Airbnb listings across major U.S. cities, including New York City, San Francisco, and Washington D.C. Each observation represents one unique property and includes 29 variables describing listing characteristics, host attributes, location details, and customer engagement metrics. Key variables include price, property_type, room_type, accommodates, bathrooms, bedrooms, beds, number_of_reviews, review_scores_rating, and policy-related factors such as cancellation_policy and instant_bookable.

The dataset also contains semi-structured text fields such as amenities, description, and neighbourhood. The amenities column provides valuable information about features offered by each listing (e.g., Wi-Fi, kitchen, air conditioning) but appears as an unstructured text list, requiring further processing before use in modeling. The price variable is right-skewed due to luxury and high-end listings, so using a log transformation helps stabilize variance and capture proportional differences more effectively.

Some variables contain missing values, particularly in bathrooms, bedrooms, and review_scores_rating, which likely stem from incomplete host entries or inactive listings. Additionally, the dataset also focuses heavily on positive features such as amenities and property details while excluding negative factors such as poor location or customer complaints. This creates a potential bias toward well-performing listings, which may slightly overstate the average quality of Airbnb properties.

Overall, the dataset provides sufficient structure and richness to explore how listing features, host attributes, and amenities influence property prices and booking competitiveness on Airbnb.

Data Cleaning:

This dataset currently contains 29 variables, including 9 categorical variables, 5 string variables, and 15 numerical variables. After reviewing the dataset, the following processing steps were performed:

1. Remove unnecessary columns**:** description, bed_type (since 98% of the dataset consists of real beds), host_has_profile_pic, name, neighbourhood, thumbnail_url.

2. property_type**:** Since apartments and houses are particularly numerous (approximately 66% apartments and 22% houses), all other property_types were categorized as "other."

3. room_type**:** There are three types: Private room, Entire home/apartment, and Shared room (this is very rare, with only 26 instances).

4. cancellation_policy**:** Since super_strict_30 and super_strict_60 are rare, all policies (strict, super_strict_30, super_strict_60) are grouped under strict.

5. host_response_rate**:** Remove the percentage sign, retaining only the numerical value.

6. amenities**:** Since the dataset combines multiple amenities into a single column, clean the data as follows: First, place over 70,000 rows of amenities into a list separated by commas. Iterate through each item, extract all unique amenities, and create a boolean column.After cleaning, 127 boolean columns were generated. To streamline subsequent analysis, only the 28 boolean columns with a total sum exceeding 10,000 were selected.

7. Encode all categorical variables (room_type, cleaning_fee, city, cancellation_policy, instant_bookable).

Methodology:

For our research question: How can Airbnb evaluate listings by analyzing factors such as location, room type, and amenities to determine appropriate and competitive prices?, we will break it down into two sub-questions:

The first one is what factors determine Airbnb pricing? We currently have numerous features like property_type, room_type, bathrooms, etc. We will analyze these using linear regression, Lasso, and linear regression with interaction to directly quantify each feature's impact on price. We will first establish a linear regression model incorporating all numerical features and encoded categorical features to develop an initial cognitive framework of price-influencing factors. Lasso will automatically identify truly significant variables through feature selection, eliminating irrelevant items. Finally, we will explore synergistic effects between different variables using linear regression with interaction terms. This phase is expected to generate a list of core drivers for enhancing property prices.

The second sub-question is how to predict a reasonable price? We will utilize the model obtained from the first question to provide an expected price for each property listing and calculate the gap between the expected price and the actual price, thereby identifying listings that are significantly undervalued or substantially overvalued. We will categorize these listings and can offer pricing recommendations to landlords. In this section, we will employ a random forest model, as it enhances the prediction accuracy of the model and captures aspects overlooked by linear models.

Modeling:

To address the business problem of determining competitive and fair Airbnb prices, several models were developed and compared to identify the best predictive approach. The target variable was the logarithm of price, which helped reduce the impact of extreme values from luxury listings and produced a more balanced distribution.

1. Multiple Linear Regression

The first model applied was an Ordinary Least Squares regression using 51 features related to location, property type, room type, capacity, amenities, host characteristics, and policy variables.

The model achieved an R^2 of 0.596 and an adjusted R^2 of 0.595, indicating that roughly 60% of price variation was explained by the features. Key positive predictors included accommodates, bedrooms, and bathrooms, while listings in Los Angeles, San Francisco, and shared or private rooms were associated with lower prices. Figure 1 confirms that residuals were randomly distributed and approximately normal, supporting the model's assumptions. While interpretable, this model assumes linearity and independence, which limits its ability to capture complex interactions.

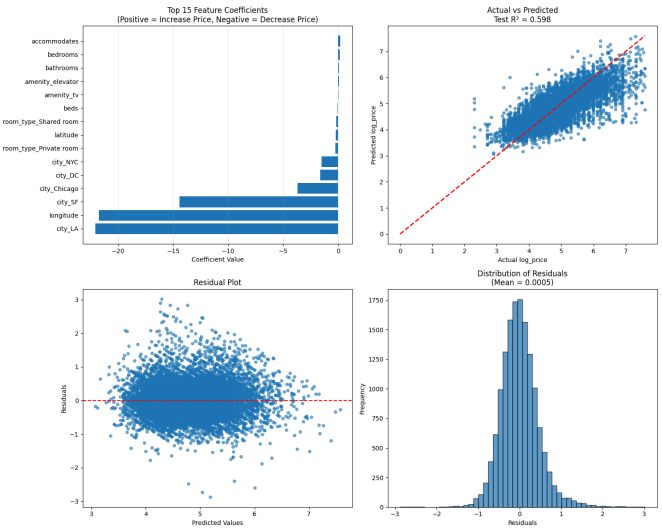

Figure 1

Figure 1. Illustrates the diagnostic results for the baseline linear regression model. The top-left plot shows the fifteen most influential coefficients, where *accommodates*, *bedrooms*, and *bathrooms* have strong positive effects on price, while listings in cities like Los Angeles or San Francisco and shared room types have strong negative impacts. The actual-versus-predicted plot (top-right) indicates a close alignment along the 45-degree line, confirming that the model predicts log-prices reasonably well with limited systematic bias. The residual plot (bottom-left) shows a random spread around zero, suggesting no major heteroskedasticity, and the residual histogram (bottom-right) is approximately normal, supporting the linear model's assumptions.

2. Lasso Regression and Feature Selection

To enhance feature selection and reduce overfitting, a Lasso regression was trained with cross-validation. The optimal regularization strength was found at alpha = 0.000408, resulting in a Test R^2 of 0.5815 and 5-fold RMSE of 0.4653. Lasso retained 48 of the 51 predictors, identifying the most influential variables while removing redundant ones. Figure 2 shows the cross-validation curve for alpha selection, and Figure 3 highlights the retained features and their coefficients. The top predictors—accommodates, bedrooms, and bathrooms—remained

consistent with OLS results. Lasso simplified the model without a significant loss in accuracy, balancing interpretability with generalization.

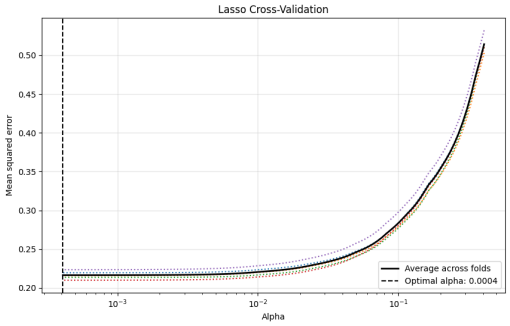

Figure 2

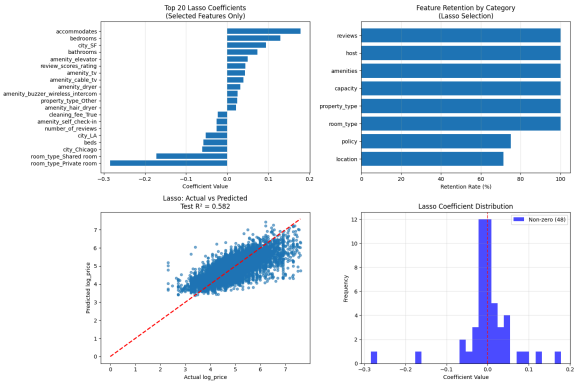

Figure 3

Figure 2. Presents the cross-validation curve used to select the optimal alpha for Lasso regularization. The minimum mean-squared-error occurs near alpha = 0.0004, balancing bias and variance effectively.

Figure 3. Visualizes the selected features and their effects. The left panel shows the top coefficients. It can be observed that accommodations and bedrooms again have the largest positive impacts, aligned with the result from linear regression, indicating that the more people it accommodates, the higher the price. Meanwhile, the city still exerts a significant influence on prices, particularly with lower prices observed in West Coast cities. Notably, SF can positively impact prices in Lasso, which contradicts linear regression. This indicates that after controlling for other variables, SF possesses a certain premium-generating capability. The feature-retention chart confirms that nearly all categories, such as host, property type, and amenities, retain predictive power. The lower panels show the predicted versus actual values and coefficient distribution, highlighting that 48 features remain non-zero after penalization, simplifying the model while maintaining accuracy.

3. Linear Regression with Interaction Terms

To capture relationships between variables that influence pricing jointly, an extended linear model with 23 interaction terms was created based on Lasso-selected features. The model achieved a Test R^2 of 0.6003, showing a modest but meaningful improvement in predictive performance. For private room types, the marginal price increase from adding a bathroom is relatively low, indicating that guests' price expectations for private rooms do not overly depend on hardware configurations. Instead, they place greater emphasis on factors such as occupancy capacity and cost-effectiveness. In the interaction between cities and amenities, both Los Angeles and Washington, D.C. exhibit negative interaction coefficients with air conditioning. This suggests that in these cities, providing air conditioning is no longer considered a premium

feature but rather a “basic standard,” leading to saturated or even negative marginal contributions to price.

Figure 4 displays the top interaction effects and performance comparison, confirming that accuracy improved slightly at the cost of moderate model complexity.

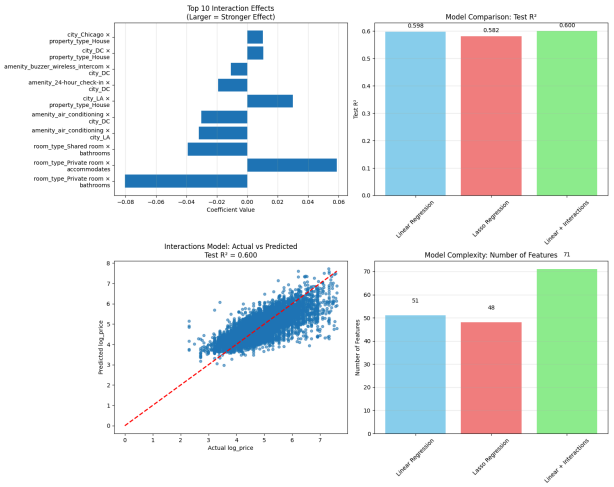

Figure 4

Figure 4. Compares the interaction-enhanced model to earlier versions. The top-left chart identifies the strongest interaction effects, including *Private room* × *bathrooms* (negative) and *Private room* × *accommodates* (positive), which reveal how price sensitivity changes when these variables combine. The top-right bar chart shows a small but meaningful improvement in predictive power compared with previous models. The bottom-left plot confirms higher alignment between predicted and actual prices, and the bottom-right chart shows that model complexity increased modestly to 71 features—an acceptable trade-off for improved performance.

4. Random Forest for Price Rationality Analysis

A Random Forest Regressor was used to capture nonlinear effects and perform a rationality assessment of listing prices. The model achieved a Test R^2 of 0.6875, outperforming all linear-based models while maintaining acceptable overfitting control (Train $R^2 = 0.8552$). Figure 5 shows that the most important features were *room type (Private room)*, *bed*rooms, and accommodates, followed by location coordinates and bathrooms. Location also demonstrates significant importance, indicating its impact on pricing. Amenities such as “TV,” “dryer,” and “elevator” rank within the top 20 factors, further highlighting customers' emphasis on supplementary facilities and their willingness to pay a premium for them.

The Random Forest model classified listings into three categories: undervalued (10%), overvalued (10%), and fairly priced (80%). Undervalued listings averaged an actual price of $74.15 versus a predicted $141.37, while overvalued properties averaged $429.17 versus a predicted $184.34. These results reveal actionable opportunities for pricing adjustment, supporting more accurate recommendations for hosts.

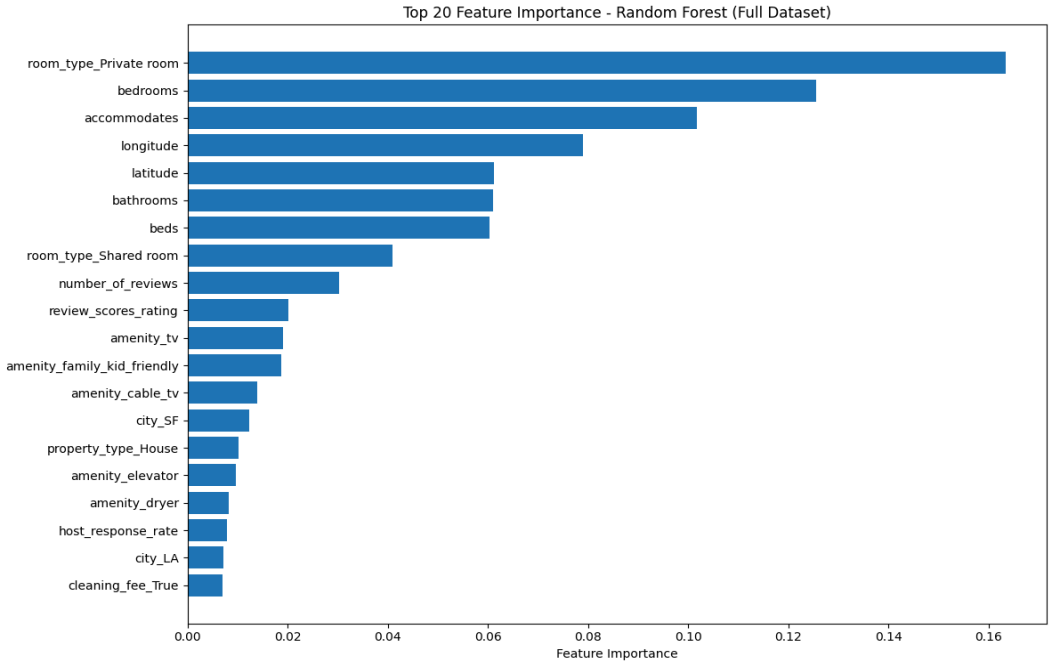

Figure 5

Figure 5. Shows the top twenty feature importances derived from the Random Forest model. The three most influential variables are *room_type = Private room*, *bedrooms*, and *accommodates*, followed by geographic coordinates and bathroom count. These results reinforce earlier findings while demonstrating that tree-based methods can capture nonlinear effects and interactions automatically. The model's feature ranking also validates that physical capacity and room type are the dominant determinants of Airbnb pricing across markets.

Each model presents a trade-off between interpretability and predictive performance. Linear and Lasso models offer clarity on how individual features affect price, which is useful for explaining pricing logic to hosts and for policy transparency. However, they are limited by their linearity assumptions and cannot easily capture complex patterns. Interaction models improve flexibility by accounting for feature relationships but increase model complexity. Random Forest captures nonlinear dependencies automatically and achieves the highest accuracy, though it is less interpretable and more computationally intensive.

Combining both linear and tree-based models provides a balanced framework: the linear models offer insights into pricing drivers, while Random Forest ensures strong predictive performance and robust price evaluations across listings.

By combining interpretability and predictive power, these models allow Airbnb to align prices more closely with listing characteristics, improve booking efficiency, and increase host earnings. The Random Forest's ability to detect under- and overvalued listings enables Airbnb to fine-tune dynamic pricing systems, recommend optimal price ranges, and maintain fairness and transparency in the marketplace. This directly addresses the business problem by improving both pricing accuracy and market trust—key dimensions of value for the firm.

Evaluation:

Model evaluation focused on assessing both predictive performance and business relevance to ensure the selected model provides actionable pricing insights for Airbnb. The primary metrics used were R^2, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), which measure how well predicted log-prices align with actual values. R^2 quantifies the proportion of price variation explained by the model, while RMSE and MAE capture the average prediction error in log-scaled terms.

Among all models tested, the Random Forest achieved the best overall performance (Test $R^2 = 0.6875$, Test RMSE = 0.4007, Test MAE = 0.2916), outperforming both linear and Lasso regression. This improvement reflects a more accurate fit across listings of varying price ranges, especially for nonlinearly distributed amenities and location effects. A five-fold cross-validation RMSE of 0.4024 confirmed consistent performance, suggesting the model generalizes well to unseen data. These results indicate that the model effectively reduces prediction error by about 10–15% compared to baseline regression methods, which translates to more accurate pricing recommendations and higher host satisfaction.

From a business standpoint, evaluation extends beyond statistical accuracy. The key measure of success is whether the model's insights can improve pricing fairness and revenue optimization. The Random Forest residual analysis identified undervalued and overvalued properties, providing clear opportunities for host guidance. If Airbnb were to integrate this model into its dynamic pricing system, the projected benefit could include higher booking rates, improved host retention, and better customer satisfaction due to perceived fairness in pricing.

Quantifying direct return on investment is difficult without access to full financial data from the platform. Still, an approximate framework can include three elements: the expected rise in bookings from better price alignment, the added revenue earned per host, and the reduction in customer churn resulting from stronger trust. A five to ten percent improvement in pricing precision could reasonably lead to a similar percentage increase in booking conversion, which at Airbnb's scale would produce a major impact on overall revenue.

Deployment:

The model can be deployed through Airbnb's host dashboard as a pricing recommendation tool. It would suggest price ranges based on property features, location, and amenities, updating automatically when listings change. Deployment should begin as an optional feature so hosts can review and adjust suggested prices. Once validated, it can evolve into a semi-automated system supported by regular model retraining to maintain accuracy.

Airbnb should ensure transparency and fairness by showing hosts how recommendations are generated and by monitoring for pricing bias across neighborhoods. Key risks include biased predictions, model drift, and host unwillingness to automation. These can be reduced through ongoing updates, fairness checks, and clear user control. This deployment would allow Airbnb to improve pricing consistency, strengthen host trust, and enhance overall market efficiency.

Appendix A: Team Member Contribution

Team Member:	**Contribution:**
Zilling Chen	Data Cleaning, modeling, PowerPoint
Chris Chen	Worked on all six sections of the write-up, PowerPoint
Sam MacArthur	Dataset selection, Evaluation, PowerPoint
Krishna Gupta	Dataset selection, PowerPoint
Neha Maqsood	Business Understanding, Formatting